COE411 – EMBEDDED AND CYBER PHYSICAL SYSTEMS

PROJECT MILESTONE II – PROJECT PROGRESS REPORT

Project Title:

Multi-Zone Environmental Monitor with Alert System

Submitted by:

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Project Overview

The original project proposed in Milestone I was an **Intelligent Fire Detection, Suppression, and Evacuation System** built around an STM32 microcontroller with full physical hardware. The system was designed to monitor a fixed direction for fire using an infrared temperature sensor, alert occupants via a buzzer and LED strip evacuation path, suppress the fire using a servo-aimed water pump, and control doors via a second servo motor to contain the fire after evacuation. Two push buttons handled evacuation confirmation and a fail-safe door override respectively.

Due to circumstances requiring a fully simulated implementation, the project has undergone two scope changes since Milestone I. First, the target microcontroller has changed from STM32 to ESP32, and the development environment is now PlatformIO with the ESP-IDF framework in VS Code, using Wokwi for hardware simulation. Second, and more significantly, several subsystems from the original design are not feasible in simulation: the water pump, the sprinkler servo aiming mechanism, and physical door control cannot be meaningfully represented in Wokwi. These have been removed from scope.

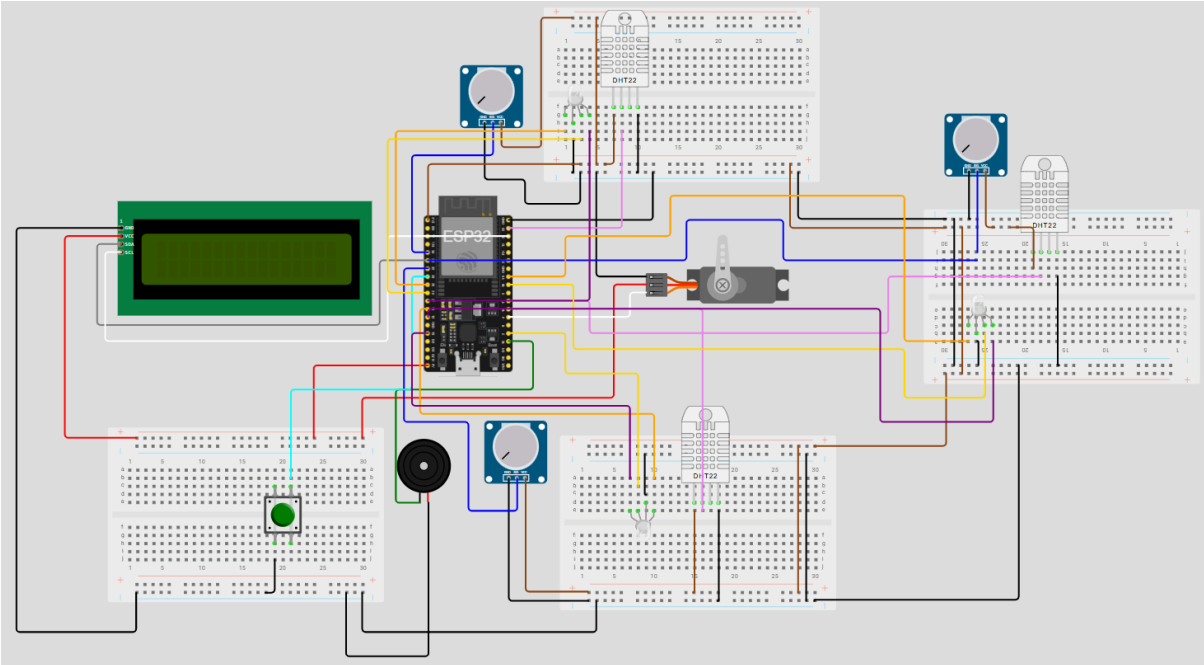
The project has been redesigned as a **Multi-Zone Environmental Monitor with Alert System**, which preserves the core embedded systems objectives of the original proposal while being fully realizable in simulation. Instead of monitoring a single fixed direction, the system now monitors three independent zones simultaneously for abnormal temperature, humidity, and gas concentration, which is a broader and arguably more realistic fire-safety scenario. The fire detection logic, threshold-based state machine, audible alarm, visual indicators, LCD status display, and human input handling from the original design are all retained and expanded. The suppression and door control subsystems are replaced by a servo motor that rotates to point at the highest-severity zone, serving as a directional alert indicator.

The two original push buttons (evacuation confirmation + fail-safe door override) have been consolidated into a single button with state-aware behavior: one press acknowledges the alert and silences the buzzer while keeping all visual indicators active, and a second press resets the system to normal once all zones have dropped below threshold. This simplification reduces hardware complexity while retaining the interrupt-driven input handling and fail-safe design philosophy of the original system.

The core embedded systems concepts demonstrated are multi-sensor ADC reading, FreeRTOS multitasking with mutex-protected shared state, interrupt-driven GPIO, PWM actuator control, and I2C communication. These are identical to what was planned in Milestone I, and in some respects more complex due to the addition of multi-zone sensor fusion and priority-based alert arbitration across three concurrent FreeRTOS tasks.

System Architecture

A. Wokwi Diagram (diagram.json in Appendix)



B. Pin Configuration Table & Components

- 1 button
- 3 potentiometer
- 3 DHT22 temperature & humidity sensors
- 3 RGB LEDs (common cathode)
- 1 16x2 LCD (I2C)
- 1 Buzzer
- 1 Servo Motor

Component	Pin	ESP32 GPIO	Protocol
DHT22 — Zone 1	SDA	GPIO 23	1-Wire (RMT)
DHT22 — Zone 2	SDA	GPIO 16	1-Wire (RMT)
DHT22 — Zone 3	SDA	GPIO 17	1-Wire (RMT)
Potentiometer — Zone 1	SIG	GPIO 34	ADC1 ch6
Potentiometer — Zone 2	SIG	GPIO 35	ADC1 ch7
Potentiometer — Zone 3	SIG	GPIO 32	ADC1 ch4
RGB LED — Zone 1	R / G / B	GPIO 25 / 26 / 27	GPIO
RGB LED — Zone 2	R / G / B	GPIO 14 / 2 / 13	GPIO
RGB LED — Zone 3	R / G / B	GPIO 19 / 18 / 12	GPIO
Servo motor	PWM	GPIO 4	LEDC PWM
Buzzer	Signal	GPIO 15	LEDC PWM
LCD 16×2 (I2C)	SDA / SCL	GPIO 21 / 22	I2C
Push button	Signal	GPIO 33	GPIO ISR

Software Architecture

A. Task Overview:
   Describe each FreeRTOS task with its priority, trigger, and purpose.

Task	**Priority**	**Trigger**	**Purpose**
vZoneSensorTask (×3)	2	Periodic, every 500 ms	Each instance is assigned to one zone. It reads the DHT22 sensor and gas-level input from the potentiometer, updates the corresponding ZoneStatus_t structure, evaluates the zone condition, and classifies it as normal, warning, or critical. After updating shared data, it sends a notification to the alert manager task through the queue.
vAlertMgrTask	3	Event-driven; unblocked by queue notifications from sensor tasks	Monitors updates from all three zones, determines the highest-priority active alert, and controls the main system outputs accordingly. It updates the servo to point toward the most severe zone, activates or silences the buzzer, and sets the RGB LEDs to reflect zone status. It also checks the system state changed by the button ISR to handle acknowledge and reset behavior.
vDisplayTask	1	Periodic, every 500 ms	Reads the latest zone data and overall system state, then formats and sends status messages to the 16×2 LCD through I2C. Its role is user feedback only, so it is assigned the lowest priority.

B. Inter-Task Communication:

Describe queues, semaphores, mutexes, or event groups used.

The system uses shared memory, a queue, and ISR-updated volatile state variables to coordinate communication between tasks safely and efficiently.

Shared data structure.
A global array, ZoneStatus_t zones[3], stores the most recent sensor readings and computed alert level for each monitored zone. Since this data is accessed by multiple tasks, all reads and writes to the array are protected by the mutex xZoneMutex. This prevents race conditions and ensures that no task reads partially updated zone information.

Queue-based notification.
Each instance of vZoneSensorTask sends a zone ID to xAlertQueue after updating its corresponding zone structure. This queue is used to notify vAlertMgrTask that new sensor data is available. As a result, the alert manager task remains blocked until fresh information arrives, which improves efficiency and avoids unnecessary polling or busy-waiting.

Volatile system state and button event flag.
The push button is handled through an interrupt service routine. Instead of performing heavy processing inside the ISR, the ISR updates shared volatile state variables such as the system state and button event flag. The alert manager task checks these values during execution to perform actions such as silencing the buzzer after acknowledgement and allowing reset only when all zones are safe. Declaring these variables as volatile ensures the compiler always reads their most recent values, even when they are modified asynchronously by the ISR.

Summary of IPC Primitives

Primitive	Between	Purpose
xZoneMutex (Mutex)	All tasks ↔ zones[3]	Protects shared zone data during read/write access
xAlertQueue (Queue)	vZoneSensorTask instances → vAlertMgrTask	Notifies the alert manager that new zone data is available

system_state / button flag (volatile)	Button ISR → vAlertMgrTask	Passes button-triggered state changes such as acknowledge and reset requests
zones_all_clear (volatile)	vAlertMgrTask → Button ISR	Indicates whether all zones are below threshold so reset is allowed safely

C. Timing Constraints and Scheduling:
Identify time-critical tasks and their deadlines.

The system includes both periodic and event-driven activities.

Time-critical functions are assigned higher priority to ensure that hazardous conditions are detected and acted upon quickly, while non-critical user-interface updates are allowed more relaxed timing.

Task / Function	**Timing Requirement**	**Notes**
vZoneSensorTask (×3)	Periodic, every 1 s	Each task samples one zone and updates its corresponding status structure. The sensor sampling period is selected to remain compatible with DHT22 timing limitations while still providing sufficiently frequent environmental monitoring.
vAlertMgrTask	Response within 100 ms of a new sensor update	This is the most time-critical software task. Once new zone data is received through the queue, it must quickly evaluate the global system condition and update the buzzer, RGB LEDs, and servo direction so that the alert is clearly visible in real time.

vDisplayTask	Periodic, every 500 ms to 1 s	LCD updates are not safety-critical, so this task has the lowest priority. A moderate refresh period is sufficient for readable user feedback without consuming unnecessary CPU time.
Button ISR	Immediate interrupt response	The button press must be detected with minimal latency. The ISR performs only lightweight processing by setting shared state or flags, while the main action is handled later by the alert manager task. This keeps interrupt latency low and avoids blocking inside the ISR.

The scheduling strategy gives highest importance to alert handling, followed by sensor monitoring, while display updates remain the lowest-priority activity. This ensures that hazardous conditions are processed promptly and that user-interface functions do not delay safety-related actions. Periodic tasks are used where deterministic sampling or display refresh is needed, while queue-based wake-up is used for responsive alert processing.

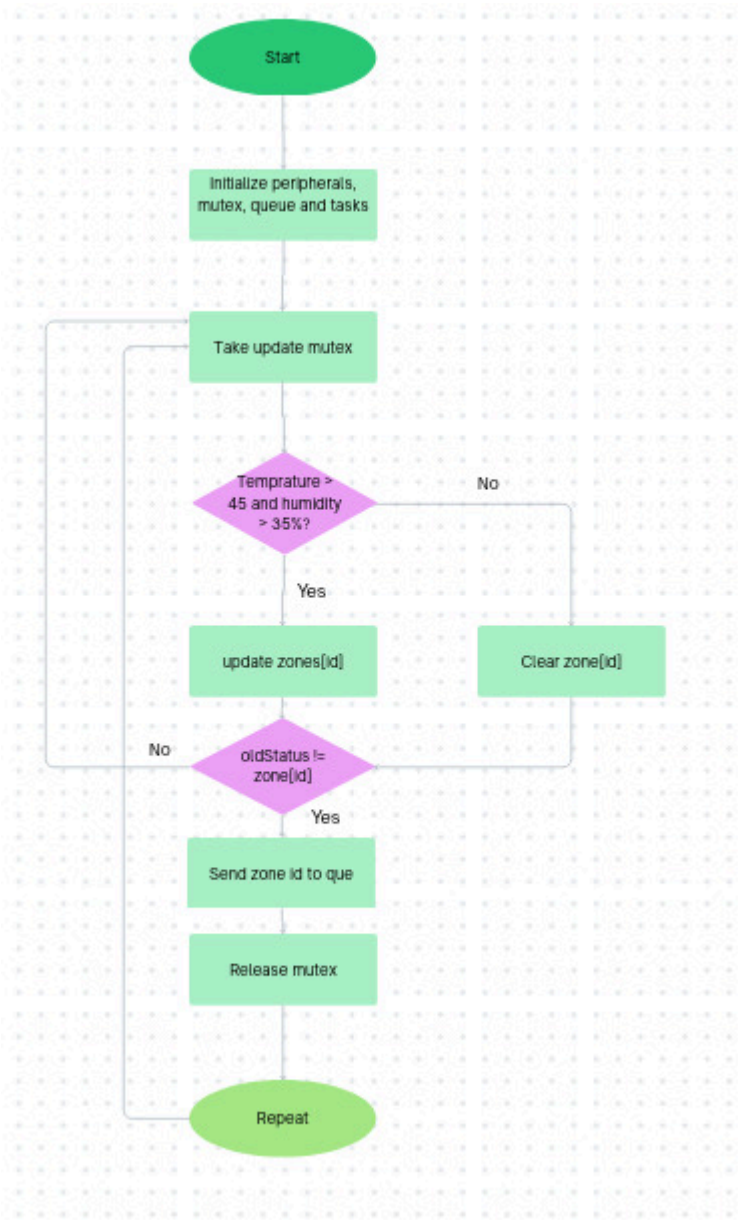
D. List project risks and mitigation strategies:

Risk Description	Impact	Likelihood	Mitigation Strategy
Sensor failure	Fire is not detected	Low probability when the sensor is brand new but increases into high probability as the sensor ages especially past the 5 year mark	Have 2 sensors in every room for redundancy
Button interrupt is blocked by a different interrupt	Doors don't open in time and the person locked inside is burned by the fires	Very high if no priority is set	Make the button interrupt a high priority interrupt

Multiple sensors detect a fire at the same time and the microcontroller can not service both in time	One of the fires is not properly contained and can break out into nearby areas	High if the fire is large	Use semaphores to ensure only 1 sensor is serviced at a time

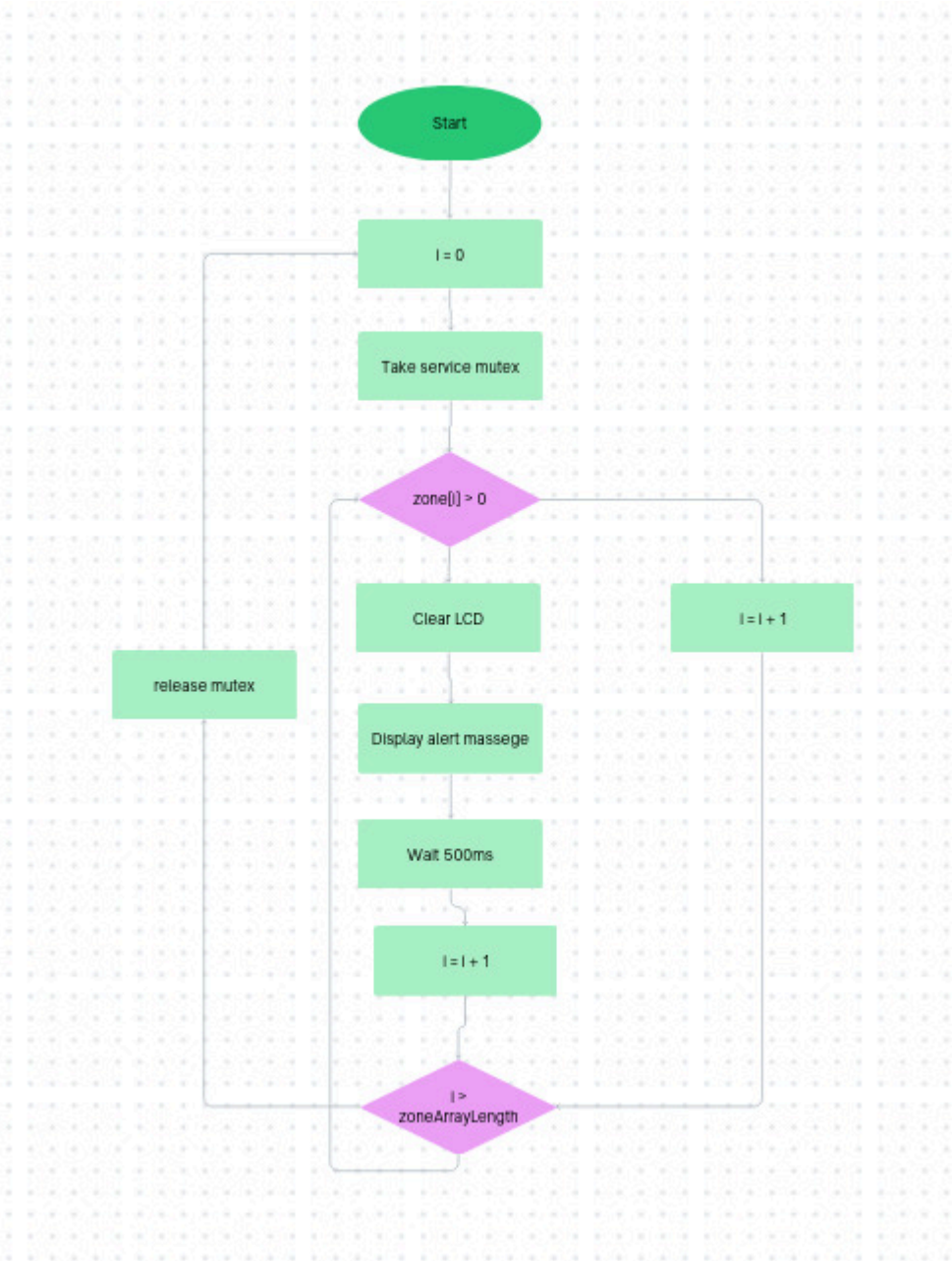
E. Attach or embed diagrams such as task interaction charts or flowcharts.

Main task:



Alert task:

LCD display task:


Start
I = 0
Take service mutex
zone[i] > 0
Clear LCD
I = I + 1
release mutex
Display alert massege
Wait 500ms
I = I + 1
I > zoneArrayLength

Current Progress and Next Steps

Discuss the tasks that have been completed and list remaining tasks with a target completion timeline.

So far we have laid out the diagram for the simulation as well as the flowcharts for the functions that will be a part of our program. We will now work on implementing the software and ensuring it is able to run without any issues by the project's deadline. The only 2 tasks left are writing the software needed to drive the microcontroller as well as testing and debugging. We will tackle this by first implementing and making sure each task can complete its job individually then making sure it is all able to synchronise correctly without any deadlocks or missed deadlines.We hope to have this completed by the 19th of April.

Appendix

diagram.json: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